SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED: CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

ANS:

A surveillance system requires a technique that provides a good balance between accuracy, speed, robustness, and computational requirements.

Technique	Speed	Accuracy	Computational Cost	Suitability
Viola-Jones	Very Fast	78%	Low	Good for strict real-time systems
YOLO	Moderate/Fast	92%	Moderate	Excellent balance
Deep Learning	Slow	95%	High	Best accuracy but unsuitable for strict latency

Comparison and Evaluation:

Viola-Jones is computationally efficient and provides very fast detection, making it suitable for systems with limited hardware and very strict latency requirements. However, its 78% accuracy is relatively low, which can result in missed or incorrect detections.

YOLO provides a much better balance between speed and accuracy. With 92% accuracy and moderate processing speed, it can perform object detection in near real time while maintaining reliable detection. It also provides better robustness than Viola-Jones for different object positions and environmental conditions.

Deep Learning provides the highest accuracy at 95% and generally offers strong robustness. However, its higher computational requirements and slower processing make it less suitable when the surveillance system has a strict latency requirement.

Decision:

YOLO should be selected for most real-time surveillance applications where latency is important but acceptable accuracy must also be maintained.

The reason is that YOLO improves accuracy from 78% to 92% compared with Viola-Jones while requiring less computation than the high-accuracy Deep Learning approach. It therefore provides the best accuracy–speed trade-off.

If the deployment uses extremely limited hardware and latency is more important than detection accuracy, Viola-Jones can be selected. On a GPU-equipped system where latency is less restrictive, Deep Learning may be preferred for maximum accuracy.

Conclusion:

For a practical surveillance system with strict latency constraints and acceptable accuracy requirements, YOLO is the most appropriate choice because it offers high accuracy, good robustness, and sufficiently fast processing without the excessive computational cost of deeper models.

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

ANS:

The robustness of the three methods can be compared by observing their detection accuracy under normal, occlusion, and low-light conditions.

Method	Normal	Occlusion	Low Light	Average Accuracy
Viola-Jones	80%	60%	55%	65.0%
YOLO	90%	75%	70%	78.3%
Deep Learning	95%	85%	80%	86.7%

Comparison and Evaluation:

Viola-Jones performs reasonably well in normal conditions with 80% accuracy, but its performance drops significantly under occlusion and low-light conditions. This shows that it has low robustness against environmental variations. Its main advantage is low computational cost and fast processing.

YOLO performs better than Viola-Jones in all three conditions. It achieves 90% accuracy in normal conditions, 75% under occlusion, and 70% in low light. Therefore, it provides better robustness while maintaining relatively fast detection, making it suitable for real-time surveillance.

Deep Learning provides the highest accuracy in every condition: 95% in normal conditions, 85% under occlusion, and 80% in low light. Its average accuracy of 86.7% is also the highest. This indicates that it is the most robust method against environmental changes.

Trade-offs:

- Viola-Jones: Fast and computationally inexpensive, but accuracy decreases greatly in difficult environments.
- YOLO: Provides a strong balance between accuracy, robustness, and computational requirements.
- Deep Learning: Gives the highest accuracy and robustness but requires greater computational resources and may have higher latency.

Conclusion:

Deep Learning provides the best overall performance across varying environmental conditions, with an average accuracy of 86.7%. It should be selected when detection reliability is the highest priority and sufficient computational resources are available.

However, if the system requires real-time performance with limited computational resources, YOLO is the better practical choice because it provides good robustness with lower computational requirements than Deep Learning.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions
- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

ANS:

Optical flow methods estimate the movement of objects between consecutive frames. Method A and Method B have different strengths and weaknesses.

Factor	Method A	Method B
Vector stability	High	Low
Small motion detection	Poor	Excellent
Noise sensitivity	Low	High
Reliability	High	Moderate
Suitability for dynamic scenes	Good	Depends on noise handling

Comparison and Evaluation:

Method A produces stable and reliable motion vectors and is less affected by noise. However, it may fail to detect small or subtle movements. It is suitable for applications where overall object movement and reliability are more important than detecting fine details.

Method B can detect fine and small motions, making it useful for applications requiring detailed motion analysis. However, its vectors are noisy, which can cause false motion detections and reduce tracking stability. It may require additional filtering or smoothing to improve reliability.

Decision:

For general motion tracking in dynamic scenes, Method A is more suitable because stable vectors provide reliable tracking even when the scene contains significant movement and noise. Consistent tracking is usually more important than detecting very small motions.

However, Method B is preferable when detecting subtle movements is critical, such as gesture recognition or precise motion analysis. In this case, noise-reduction techniques such as filtering can be applied before tracking.

Conclusion:

Method A is the better choice for reliable motion tracking in dynamic scenes because it provides stable vectors and is less sensitive to noise. Method B offers better fine-motion detection but requires additional noise-handling techniques. Therefore, the final choice should depend on whether the application prioritizes tracking stability or sensitivity to small movements.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

ANS:

A Gaussian Mixture Model (GMM) is commonly used for background subtraction by modeling the distribution of pixel values over time. Its performance varies depending on the type of background.

Comparison and Evaluation:

In a dynamic background, elements such as moving trees, waving leaves, rain, or water can continuously change pixel values. GMM may interpret these changes as foreground objects, resulting in high false positives.

In a static background, the pixel distribution is more stable, so GMM can model the background effectively. However, if an object remains stationary for a long time, GMM may gradually learn it as part of the background. This can cause missed detections.

Therefore, GMM alone is not sufficient for reliable detection in all environmental conditions.

Scenario	GMM Performance	Main Problem
Dynamic background	Poor	High false positives
Static background	Better, but may miss objects	Missed detections

Proposed Improvements:

- Adaptive learning rate: Adjust the background update rate according to scene changes. A slower update can prevent stationary objects from being quickly absorbed into the background.
- Morphological filtering: Use erosion and dilation to remove small noise and fill gaps in detected objects.
- Shadow detection: Remove shadows from the foreground mask to reduce false positives.
- Temporal and spatial filtering: Analyze motion across multiple frames to distinguish actual objects from temporary background changes.
- Combine GMM with optical flow: Optical flow can provide motion information and help distinguish real moving objects from dynamic background noise.
- Use a deep-learning detector: YOLO or another object detector can be combined with GMM to improve detection reliability in complex scenes.

Conclusion:

GMM alone is not sufficient because it can produce false positives in dynamic backgrounds and miss stationary objects in static backgrounds. An adaptive GMM combined with morphological filtering and optical flow would provide better robustness. For highly complex surveillance environments, combining GMM with a deep-learning object detector would provide the most reliable results, although it increases computational cost.

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

ANS:

The two methods provide a trade-off between accuracy and computational efficiency.

Factor	Method A	Method B
Error	5% (Low)	10% (Moderate)
Accuracy	Higher	Lower
Computational cost	High	Low
Processing speed	Slower	Faster
Real-time suitability	Lower	High
Offline suitability	High	Moderate

Comparison and Evaluation:

Method A has only 5% error, so it provides more accurate motion estimation. However, its high computational requirement makes it slower and may cause delays when processing frames in real time. Therefore, it is better suited for applications where accuracy is more important than processing speed.

Method B has a higher 10% error, but its low computational cost allows faster processing. This makes it more suitable for real-time applications where low latency and quick response are important.

Application-Based Decision:

- Real-time applications: Method B is preferred because its low computational cost allows faster motion estimation and continuous processing. The additional 5% error may be acceptable when immediate response is more important.
- Offline applications: Method A is preferred because there is no strict latency requirement. Its higher computational cost can be tolerated, while its lower error provides more accurate results.

Conclusion:

The choice depends on the application requirements. Method B is better for real-time systems because of its low computation and faster processing, while Method A is better for offline systems because its lower error provides higher accuracy. Thus, the decision involves a trade-off between accuracy and computational efficiency.

Code of Conduct:

I G.Joel Richard (192524054) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand